

# AIMO Proof Pilot — Final Grade

Problem (#\_ID): 1\_DIVALL

Submission: model\_5

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Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

## 1. Final mark

Final mark (choose one):

## 2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

A complete, correct solution. It proves the common residue modulo  $G$ , shows this residue is nonzero, rules out all  $k \geq 4$ , and reduces the  $k = 3$  case to  $G = (c^2 + 1)/m$  with  $m \in \{1, 2\}$ , eliminating both under the assumption  $G > 998285$ . It then gives a valid construction attaining 998285. No markscheme-level errors. Scores **7**.